

probability

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1 Counting

1. How many ways are there to permute the letters in the word MISSISSIPPI?

There are 11 letters in MISSISSIPPI. If each letter is labeled $M_1, I_1, S_1, S_2, I_2, S_3, S_4, I_3, P_1, P_2, I_4$, there are $11!$ permutations.

However, we will overcount, as the same letters are not actually distinct. The number of times each distinct permutation of MISSISSIPPI is overcounted is $1! \cdot 4! \cdot 4! \cdot 2!$, since there is 1 M, 4 I's, 4 S's, and 2 P's. Thus, our answer is $\frac{11!}{4!4!2!} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 2} = \frac{11 \cdot 10 \cdot 9 \cdot 7 \cdot 5}{1} = 34650$.

2. (a) How many 7-digit phone numbers are possible, assuming the first digit can't be a 0 or a 1?

For the first digit, we have 8 options. Every other digit has 10 options. Our answer is $8 \cdot 10^6$.

- (b) Re-solve (a), except now assume also that the phone number is not allowed to start with 911 (since this is reserved for emergency use, and it would not be desirable for the system to wait to see whether more digits were going to be dialed after someone has dialed 911.)

We will take the answer from the previous part, but we must remove all the phone numbers that begin with 911. Since there are 10 options for each of the 4 remaining, there are 10^4 such bad phone numbers. Thus, our answer is 7960000.

3. Fred is planning to go out to dinner each night, Monday through Friday, with each dinner being at one of his ten favorite restaurants.

- (a) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is not willing to eat at the same restaurant more than once?

There are $\binom{10}{5}$ combinations of restaurants to eat at. Since the order matters, our answer is $\binom{10}{5} \cdot 5! = \frac{10!}{5!} = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240$.

- (b) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is willing to eat at the same restaurant more than once, but is not willing to eat at the same place twice in a row (or more)?

Note that on Monday, Fred can eat at any of the 10 restaurants. On every following day, however, he cannot eat at the same place as the day before, so he has 9 options. Thus, the answer is $10 \cdot 9^4$.

4. A *round-robin* tournament is being held with n tennis players; this means that every player will play against every other player exactly once.

- (a) How many possible outcomes are there for the tournament (the outcome lists out who won and lost for each game)

Since everybody plays each other regardless of any match outcomes, the total number of outcomes is 2^x , where x is the number of matches played. Since there are n players, there are $\binom{n}{2} = \frac{n(n-1)}{2}$ matches, so our answer is $2^{\frac{n(n-1)}{2}}$.

- (b) How many games are played in total?

$$\frac{n(n-1)}{2}$$

5. A *knock-out tournament* is being held with 2^n tennis players. This means that for each round, the winners move on to the next round and the losers are eliminated, until only one person remains. For example, if initially there are $2^4 = 16$ players, then there are 8 games in the first round, then the 8 winners move on to round 2, then the 4 winners move on to round 3, then the two winners move on to round 4, the winner of which is declared the winner of the tournament. (There are various systems for determining who plays whom within a round, but these do not matter for this problem.)

- (a) How many rounds are there?

Each round cuts the number of players in half, until we get to 1. If we divide 2^n by 2 n times, we get 1. Thus, there will be n rounds in the tournament.

- (b) Count how many games in total are played, by adding up the number of games played in each round.

In round 1, 2^{n-1} games are played. In round 2, 2^{n-2} games are played. Each round, the number of games is cut in half. We can see that the number of games played in total is thus $2^{n-1} + 2^{n-2} + \dots + 2^0$, which equals $2^n - 1$.

- (c) Count how many games in total are played, this time by directly thinking about it without doing almost any calculation.

Since we have 2^n players, and each round knocks out a player permanently, to end up with one winner we will need to eliminate $2^n - 1$ players. $2^n - 1$ matches will be played.

6. There are 20 people at a chess club on a certain day. They each find opponents and start playing. How many possibilities are there for how they are matched up, assuming that in each game it *does* matter who has the white pieces (in a chess game, one player has the white pieces and the other has the black pieces)?

Let's number the players from 1...20. We generate matchups by repeatedly taking the player with the lowest number from the unmatched pool, then picking another player who is unmatched and pairing them. Any valid way to generate matchups can be written as a sequence s of 10 numbers, where s_i is the other player matched with the player with the lowest number in the i th matchup.

If two valid sequences s and s' are different, then the matchups are different too, as if we consider the first i such that $s_i \neq s'_i$, both of the i th matchups will include the same 'smallest-number' player, but

they will be matched with a different opponent.

Thus, counting the number of such sequences is equivalent to the number of different ways to match the players. Initially, there are 19 choices for s_1 , then 17 choices for s_2 , and so long. Thus, the number of ways to just pair the players is $19 \cdot 17 \cdot 15 \cdot 13 \cdot 11 \cdot 9 \cdot 7 \cdot 5 \cdot 3$.

However, for every way to pair the players, we have to assign players the white and black pieces. Since there are 10 pairings, and 2 colorings for each pairing, our final answer is $2^{10} \cdot 19 \cdot 17 \cdot 15 \cdot 13 \cdot 11 \cdot 9 \cdot 7 \cdot 5 \cdot 3$.

7. Two chess players, A and B, are going to play 7 games. Each game has three possible outcomes: a win for A (which is a loss for B), a draw (tie), and a loss for A (which is a win for B). A win is worth 1 point, a draw is worth 0.5 points, and a loss is worth 0 points.

- (a) How many possible outcomes for the individual games are there, such that overall player A ends up with 3 wins, 2 draws, and 2 losses?

Out of 7 games, we must select 3 for A to win, which is $\binom{7}{3}$. Then out of the remaining 4, we need to select two draws, so our final answer is $\binom{7}{3} \binom{4}{2} = 210$.

- (b) How many possible outcomes for the individual games are there, such that A ends up with 4 points and B ends up with 3 points?

If we know how many draws there are, we can calculate how many wins/losses from A are needed in the remaining games. The possible numbers of draws are 0, 2, 4, and 6. If there are an odd number of draws, then neither player will have an integral number of points.

For 0 draws, A wins four and loses the rest, so the answer is $\binom{7}{3} = 35$.

For 2 draws, A needs 3 wins and 2 losses to round out the required number of points for A and B, so we have $\binom{7}{2} \cdot \binom{5}{3} = 210$.

For 4 draws, A needs 2 wins and 1 loss. Thus, the number of ways is $\binom{7}{4} \cdot \binom{3}{2} = 105$.

For 6 draws, A needs to win the other game. This gives us 7 ways.

Our answer is the sum of all: $35 + 210 + 105 + 7 = 357$

- (c) Now assume they are playing a best-of-7 match, where the match will end when either player has 4 points or when 7 games have been played, whichever is first. For example, if after 6 games the score is 4 to 2 in the favor of A, then A wins the match and they don't play a 7th game. How many possible outcomes for the individual games are there, such that the match lasts 7 games and A wins by a score of 4 to 3?

We must subtract from our answer to the previous part all ways which player A wins too early, in less than 7 games.

For 0 draws, any situation where A wins 4 games prior to the seventh match must be subtracted. We must subtract $\binom{6}{4}$ from 35, giving $35 - 15 = 20$.

For 2 draws, if A obtains 2 draws and 3 wins before the seventh game, the match will end early. We must subtract $\binom{6}{2} \cdot \binom{4}{3}$ from our original value of 210 to get $210 - 60 = 150$.

For 4 draws, we must subtract any scenario where A gets 4 draws and 2 wins before the loss in the seventh game, meaning our answer is $105 - \binom{6}{4} = 105 - 15 = 90$.

Lastly, for 6 draws, if A wins before the seventh game, then after game six A will have 2.5 points from draws and 1 point from winning. Thus, A cannot achieve 4 points prematurely, meaning all 7 outcomes with 6 draws and 1 win from A are still valid.

Summing these up we get $20 + 150 + 90 + 7 = 267$.

8. (a) How many ways are there to split a dozen people into 3 teams, where one team has 2 people, and the other two teams have 5 people each?

We can first choose 5 people, and then choose another 5, giving $\binom{12}{5} \cdot \binom{7}{5} = 16632$.

However, the order which we choose the groups of 5 shouldn't matter, so we divide our answer by two, giving 8316.

- (b) How many ways are there to split a dozen people into 3 teams, where each team has 4 people?

We first choose 4, then another 4, giving $\binom{12}{4} \cdot \binom{8}{4} = 34650$.

However, we count every possible split $3!$ times, so the answer is $\frac{34650}{6} = 5775$.

9. (a) How many paths are there from the point $(0,0)$ to $(210,211)$, where each step either consists of going one unit up or one unit to the right?

We must make $210 + 211 = 421$ moves in total, 210 of which have to be rightwards movements, and the other 211 upwards. So, the answer is $\binom{421}{210}$.

- (b) How many paths are there from $(0,0)$ to $(210,211)$ where each step consists of going one unit up or one unit to the right, and the path has to go through $(110,111)$?

We will multiply the number of paths from $(0,0)$ to $(110,111)$ by the number of paths from $(110,111)$ to $(210,211)$.

From $(0,0)$ to $(110,111)$, we make 221 steps and 110 must be to the right, which is $\binom{221}{110}$. From $(110,111)$ to $(210,211)$, we make 200 steps, 100 of which must be to the right, yielding $\binom{200}{100}$. Thus, our final answer is $\binom{221}{110} \cdot \binom{200}{100}$.

10. To fulfill the requirements for a certain degree, a student can choose to take any 7 out of a list of 20 courses, with the constraint that at least 1 of the 7 courses must be a statistics course. Suppose that 5 of the 20 courses are statistics courses.

- (a) How many choices are there for which 7 courses to take?

We will count the number of ways to choose 7 courses of the 20, but we must subtract the ways which do not involve a statistics course. Thus, our answer is $\binom{20}{7} - \binom{15}{7}$.

- (b) Explain intuitively why the answer to (a) is *not* $\binom{5}{1} \cdot \binom{19}{6}$

The above expression counts the number of ways to select a statistic course and multiplies it by the number of ways to choose 6 of the remaining courses. However, this method of counting will overcount. For any valid selection of courses, our method of counting counts it at least once. If we choose 2 statistics courses however, note that there are two ways to count the final selection, depending on which course is selected from the initial $\binom{5}{1}$ choice.

For example, if we label the 15 non-statistics courses $c_1 \dots c_{15}$ and label the 5 statistics courses $s_1 \dots s_5$, a selection of $s_1, s_2, c_1, c_2, c_3, c_4, c_5$ is counted twice: We could first pick s_1 , then the combination $\{s_2, c_1, c_2, c_3, c_4, c_5\}$ from the remaining 19, or we could pick s_2 , then choose $\{s_1, c_1, c_2, c_3, c_4, c_5\}$ from the remaining.

11. Let A and B be sets with $|A| = n, |B| = m$.

- (a) How many functions are there from A to B (i.e., functions with domain A , assigning an element of B to each element of A)?

For each element of x A , there are m choices for $f(x)$. Thus, the answer is m^n .

- (b) How many one-to-one functions are there from A to B ? (See section A.2.1 of the math appendix for information about one-to-one functions.)

We require that $f(x) = f(y)$ implies $x = y$. Essentially, of the m elements in B , we must select n of them to be our designated outputs. We then have to permute the selected outputs and assign them to each element of a . Thus, the answer is $\binom{m}{n} \cdot n!$.

12. Four players, named A,B,C, and D, are playing a card game. A standard, well-shuffled deck of cards is dealt to the players (so each player receives a 13-card hand).

- (a) How many possibilities are there for the hand that player A will get? (Within a hand, the order in which cards were received doesn't matter.)

The answer is of course $\binom{52}{13}$.

- (b) How many possibilities are there overall for what hands everyone will get, assuming that it matters which player gets which hand, but not the order of cards within a hand?

We first choose 13 cards for A, then 13 cards for B, then 13 cards for C. This is $\binom{52}{13} \cdot \binom{39}{13} \cdot \binom{26}{13}$.

- (c) Explain intuitively why the answer to Part (b) is not the fourth power of the answer to Part (a).

Raising $\binom{52}{13}$ to the fourth power does not account for the fact that, if one player has the Ace of Spades, no other player can have this card. We must remind ourselves that we are choosing without replacement in this situation.

13. A certain casino uses 10 standard decks of cards mixed together into one big deck, which we will call a *superdeck*. Thus, the superdeck has $52 \cdot 10 = 520$ cards, with 10 copies of each card. How many different 10-card hands can be dealt from this superdeck? The order of the cards does not matter, nor does it matter which of the original 10 decks the cards came from. Express your answer as a binomial

coefficient.
Hint: Bose-Einstein

Let us number each kind of card from $1 \dots 52$. Let x_i be the number of cards i in the hand. We want the number of integer solutions to $\sum_{i=1}^{52} x_i = 10$. We want to put 10 indistinguishable items into 52 distinguishable boxes (the kinds of cards), where boxes can be empty. Using stars and bars, this gives us $\binom{52-1+10}{10} = \binom{61}{10}$.

14. You are ordering two pizzas. A pizza can be small, medium, large, or extra-large, with any combination of 8 possible toppings (getting no toppings is allowed, as is getting all 8). How many possibilities are there for your two pizzas?

Let's count the number of ways to have 1 pizza. For the toppings, we have 2^8 possible combinations. We then have 4 possible sizes, so our answer is $4(2^8) = 2^{10}$.

Thus, the number of ways to order two pizzas if the order matters is $(2^{10})^2 = 2^{20}$. However, we will double count getting pizza a then b as different than getting pizza b then a . We can't divide by two yet, since getting the same pizza twice is counted once. Thus, we first add the number of ways to order the same pizza twice, which is 2^{10} of course. This gives us $2^{20} + 2^{10}$.

We divide the answer by two so each combination of pizzas is counted once: $2^{19} + 2^9 = 524800$.

2 Story Proofs

15. Give a story proof that $\sum_{k=0}^n \binom{n}{k} = 2^n$.

The LHS describes the number of ways to choose some number of elements from a set of size n . It does this by summing the number of ways to choose k elements for all k .

The RHS describes the same thing, as picking a subset of any size from a group of n can be thought of as deciding for each item whether or not you want to include it in your subset. For each item, there is two choices: to take it, or to not take it. Thus, the total number of ways is 2^n .

16. Show that for all positive integers n and k with $n \geq k$,

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

doing this in two ways: (a) algebraically and (b) with a story, giving an interpretation for why both sides count the same thing.

Hint for the story proof: Imagine an organization consisting of $n+1$ people, with one of them pre-designated as the president of the organization.

(a) Algebraic

We use the formula for binomial coefficients: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!}$$

Note that $k > 0$, so $k - 1 \geq 0$.

$$\begin{aligned}
& \frac{n!}{k!(n-k)!} \cdot \frac{n-k+1}{n-k+1} + \frac{n!}{(k-1)!(n-k+1)!} \cdot \frac{k}{k} \\
& \frac{(n-k+1)n! + kn!}{k!(n-k+1)!} \\
& \frac{(n-k+1+k)n!}{k!(n-k+1)!} \\
& \frac{(n+1)n!}{k!(n-k+1)!} \\
& \frac{(n+1)!}{k!(n-k+1)!} \\
& \binom{n+1}{k}
\end{aligned}$$

(b) Story

Imagine a group of $n+1$ people numbered $1 \dots n+1$. We would like to select k members from this group.

One way to select k members is to add the number of groups that contain person 1 with the number of groups that don't contain person 1. The number of groups with 1 in them is $\binom{n}{k-1}$, since we choose from $2 \dots n+1$ $k-1$ more people. The number of groups that don't contain person 1 is $\binom{n-1}{k}$, since we choose k from $2 \dots n+1$. This gives $\binom{n}{k-1} + \binom{n-1}{k}$, which is the LHS.

The easy way to do this is of course $\binom{n+1}{k}$, the RHS. Thus, we show that both sides of the equation count the number of ways to choose k people from a group of n .

17. Give a story proof that

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$

for all positive integers n .

Say Jonathan has two kids, Alex and Gavin. He will get them each a pizza. A pizza can have any combination of n toppings. However, Jonathan wants to be fair, so both kids will get the same number of toppings. The number of ways to get the kids pizzas is exactly the LHS:

$$\sum_{k=0}^n \binom{n}{k}^2$$

since if k is the number of toppings the children will get, the number of ways both children can get k toppings on their pizzas is $\binom{n}{k}^2$.

Note that choosing k toppings for a pizza is the same as choosing $n-k$ toppings to not go on a pizza. Meaning, if Alex's pizza has k toppings, there are $n-k$ toppings not present on Gavin's pizza. Consider the following set of events:

$$a_1, a_2, \dots, a_n, g'_1, g'_2, \dots, g'_n$$

This set has size $2n$. Consider selecting a combination of size n from this set. If we select a_i , we put topping i on Alex's pizza. If we don't select g'_i , we put topping i on Gavin's pizza.

Note that every possible way to give the children pizzas corresponds to exactly one size n subset of the above set. A way to give them pizzas corresponds to the set of k toppings Alex got and $n - k$ toppings Gavin didn't get, which is size n . Any subset of size n from the above set yields a unique way to give the children pizzas, too. Thus, we are confident that the number of ways of selecting n elements from this set does indeed coincide with the number of ways to give Alex and Gavin pizzas.

Thus, the number of ways to buy the kids pizzas is $\binom{2n}{n}$, which happens to be the RHS of the equation.

18. Give a story proof that

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

for all positive integers n .

Hint: Consider choosing a committee of size n from two groups of size n each, where only one of the two groups has people eligible to become the chair of the committee.

Consider our solution to question 17. Imagine the same scenario, but Jonathan recognizes that Alex likes to eat more than Gavin. He will continue to ensure both boys get the same number of toppings, but now he will order extra for exactly one of Alex's toppings. Note that now, both kids must have at least one topping on their pizzas.

This new value is

$$\sum_{k=1}^n k \binom{n}{k}^2$$

because if we have k toppings (we need at least one topping), the number of ways to choose the toppings for both Alex and Gavin is $\binom{n}{k}$, but we multiply by k because Alex then gets to choose one topping to get extra of. This is the LHS of the equation.

Another way to count the number of ways to buy pizzas is as follows: First, Alex picks a topping i . There are n ways to do this. Now, consider the set

$$\{a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n, g'_1, \dots, g'_n\}$$

Note that this is the same set from the previous problem, except a_i is not present, as Alex has already chosen to get topping i with extra. Thus, we have a new set of size $2n - 1$. If we select $n - 1$ elements, then if we combine this subset with a_i , we will have a set of size n giving us a way to order the two pizzas. Thus, every way to choose an extra topping and then choose $n - 1$ from the resulting set of size $2n - 1$ yields a unique way to buy pizzas.

The number of ways is $n \binom{2n-1}{n-1}$, since there are n choices for the extra topping and then we must choose $n - 1$ from the set of $a_1, \dots, a_n, g'_1, \dots, g'_n$ with a_i removed. This is exactly the RHS of the equation. Thus, both quantities are identical for positive n , since they count the same thing.

19. Give a story proof that

$$\sum_{k=2}^n \binom{k}{2} \binom{n-k+2}{2} = \binom{n+3}{5}$$

for all integers $n \geq 2$.

Hint: Consider the middle number in a subset of $\{1, 2, \dots, n+3\}$ of size 5.

If we consider choosing 5 numbers from $1 \dots, n+3$, there are $\binom{n+3}{5}$ ways, which is the RHS.

Another way to do this is as follows: Let $k+1$ be the middle of our 5 numbers. We want to count the number of ways to choose 5 such that $k+1$ is the middle. Clearly, $k \geq 2$ and $k \leq n$, otherwise there will not be enough numbers available smaller or bigger than $k+1$.

Since $k+1$ is the middle, there are two other numbers smaller than or equal to k , and 2 numbers greater than $k+1$. The number of ways to choose the two smaller numbers is $\binom{k}{2}$, and the number of ways to choose the 2 greater numbers is picking 2 from $n+3-(k+1) = n-k+2$ which is $\binom{n-k+2}{2}$. Thus, multiplying them together gives $\binom{k}{2} \cdot \binom{n-k+2}{2}$. Summing over all valid k yields $\sum_{k=2}^n \binom{k}{2} \cdot \binom{n-k+2}{2}$, which is exactly the LHS.

20. (a) Show using a story proof that

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \dots + \binom{n}{k} = \binom{n+1}{k+1},$$

where n and k are positive integers with $n \geq k$. This is called the *hockey stick identity*.

Hint: Imagine arranging a group of people by age, and then think about the oldest person in a chosen subgroup.

Say we want to pick $k+1$ numbers from $1 \dots n+1$. This is simply $\binom{n+1}{k+1}$, the right hand side of the expression.

Another way to count this is as follows: let x be the largest number you select. To count the number of ways to pick $k+1$ elements with x as the largest, we need to select the k smaller elements, so $\binom{x-1}{k}$. Note that x must be greater than k for there to exist k smaller elements than x . Thus, we have $\sum_{x=k+1}^{n+1} \binom{x-1}{k} = \binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \dots + \binom{n}{k}$, which is the LHS of the equation.

- (b) Suppose that a large pack of Haribo gummi bears can have anywhere between 30 and 50 gummi bears. There are 5 delicious flavors: pineapple (clear), raspberry (red), orange (orange), strawberry (green, mysteriously), and lemon (yellow). There are 0 non-delicious flavors. How many possibilities are there for the composition of such a pack of gummi bears? You can leave your answer in terms of a couple of binomial coefficients, but not a sum of lots of binomial coefficients.

Let there be $30 \leq k \leq 50$ gummi bears in the pack. We have k indistinguishable items which we must put in 5 distinguishable boxes which can be empty (the flavors). With stars and bars, this is equal to $\binom{k+4}{4}$. We would like to sum this over all k , giving

$$\sum_{k=30}^{50} \binom{k+4}{4}$$

$$\sum_{k=34}^{54} \binom{k}{4}$$

According to the Hockey stick identity, we know

$$\sum_{k=4}^{54} \binom{k}{4} = \binom{55}{5}$$

Furthermore, we know that

$$\sum_{k=4}^{33} \binom{k}{4} = \binom{34}{5}$$

and

$$\sum_{k=4}^{54} \binom{k}{4} - \sum_{k=4}^{33} \binom{k}{4} = \sum_{k=34}^{54} \binom{k}{4}$$

Thus,

$$\sum_{k=34}^{54} = \binom{55}{5} - \binom{34}{5}$$

which is our answer.

21. Define $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ as the number of ways to partition $\{1, 2, \dots, n\}$ into k nonempty subsets, or the number of ways to have n students split up into k groups such that each group has at least one student. For example, $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = 7$ because we have the following possibilities:

- $\{1\}, \{2, 3, 4\}$
- $\{2\}, \{1, 3, 4\}$
- $\{3\}, \{1, 2, 4\}$
- $\{4\}, \{1, 2, 3\}$
- $\{1, 2\}, \{3, 4\}$
- $\{1, 3\}, \{2, 4\}$
- $\{1, 4\}, \{2, 3\}$

Prove the following identities:

(a)

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}.$$

Hint: I'm either in a group by myself or I'm not.

Consider a group of $n+1$ students labeled $1 \dots n+1$. There are $\left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\}$ ways to group the other kids into $k-1$ groups if student 1 is solo. If student 1 is not in their own group, then the number of ways is $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$, but we must choose which group for student 1 to be in, so we multiply by k .

Summing the two gives $\left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$

(b)

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}$$

Hint: First decide how many people are not going to be in a group.

Once again, let there be $n+1$ students labeled $1 \dots n+1$. There are $\left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}$ ways to group them into $k+1$ groups, which is the RHS.

Another way to group the students into $k+1$ groups is as follows. Let there be j students not in student 1's group. There are $\binom{n}{j}$ ways to do this. We must then split the j students into k groups, giving $\binom{n}{j} \cdot \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$ ways. If we sum this over all j , we will have the number of ways to split $n+1$ students into $k+1$ groups. j must be at least k , but no more than n , so this value is $\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$, which is the LHS of the equation.

22. The Dutch mathematician R.J. Stroeker remarked:

Every beginning student of number theory surely must have marveled at the miraculous fact that for each natural number n the sum of the first n positive consecutive cubes is a perfect square. [26]

Furthermore, it is the square of the sum of the first n positive integers! That is,

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$$

Usually, this identity is proven by induction, but that does not give much insight into why the result is true, nor does it help much if we wanted to compute the left-hand side but didn't already know this result. In this problem, you will give a story proof of this identity.

(a) Give a story proof of the identity

$$1 + 2 + \cdots + n = \binom{n+1}{2}$$

Hint: Consider a round-robin tournament (see Exercise 4)

Consider a group of $n+1$ people. If we want to choose two of them, there are $\binom{n+1}{2}$ ways to choose a pair, which is the RHS.

However, if we label this group $1 \dots n$, we can also count for each $1 \leq i \leq n$ how many ways there are to pair i with someone with a greater label. This counts the number of pairings, as each pairing has a smaller element, and each way to pair i with a bigger label is a unique pairing.

As such, for each i , there are $n+1-i$ pairings where i has the smaller label. This gives us a total number of pairings of $\sum_{i=1}^{n+1} n+1-i = \sum_{i=1}^n i = 1 + 2 + \cdots + n$, which is the LHS.

(b) Give a story proof of the identity

$$1^3 + 2^3 + \cdots + n^3 = 6\binom{n+1}{4} + 6\binom{n+1}{3} + \binom{n+1}{2}.$$

It is then just basic algebra (not required for this problem) to check that the square of the right-hand side in (a) is the right-hand side in (b).

Hint: Imagine choosing a number between 1 and n and then choosing 3 numbers between 0 and n smaller than the original number, with replacement. Then consider cases based on how many numbers were chosen.

Lets say we choose a number x from $1 \dots n+1$. We then choose a tuple (a, b, c) such that a, b, c are integers and $1 \leq a, b, c < x$. For each selection of x , there are $(x-1)^3$ tuples available, since there are $x-1$ options for all of a, b, c . Thus, the number of ways to choose a number x from $1 \dots n+1$ and then a tuple (a, b, c) following the above rule is $\sum_{x=1}^{n+1} (x-1)^3 = 1^3 + 2^3 + \cdots + n^3$, which is the LHS of our equation.

Consider how many unique elements are in the tuple. It is either three, two, or one. Let us perform casework on the number of unique elements in the tuple:

In the first case, the tuple has three distinct elements. We know that x is different than all these elements, since x is greater than them all. Thus, this is the same as choosing 4 numbers from $n+1$, then permuting the smaller three. This gives $6\binom{n+1}{4}$.

In the second case, the tuple has two distinct elements. There are $\binom{n+1}{3}$ ways to choose x and then the two numbers in the tuple. Given a, b , there are six tuples that we can make where one of them appears twice: $(a, a, b), (a, b, a), (a, b, b), (b, a, a), (b, a, b), (b, b, a)$. Thus, there are $6\binom{n+1}{3}$ choices for x and the tuple such that there are two distinct elements in the tuple.

In the final case, the tuple consists of one point. Of course, there are $\binom{n+1}{2}$ ways, since x and the tuple are uniquely defined by any two numbers chosen.

Thus, the answer is $6\binom{n+1}{4} + 6\binom{n+1}{3} + \binom{n+1}{2}$, which is the RHS of the equation.

3 Naive definition of probability

23. Three people get into an empty elevator at the first floor of a building that has 10 floors. Each presses the button for their desired floor (unless one of the others has pressed that button). Assume that they are equally likely to want to go to floors 2 through 10 (independently of each other). What is the probability that the buttons for three consecutive floors are pressed?

We can think of as the buttons people attempt to press as a tuple (a, b, c) where $2 \leq a, b, c \leq 10$. This means there are 9^3 total ways for them to attempt buttons. There are only 6 combinations that result in three consecutive floors starting from 2 and ending at 7. Thus, because there are $3! = 6$ ways to permute the three floors, there are 42 different tuples that result on three consecutive floors.

Our answer is thus $\frac{42}{9^3} = \frac{42}{729} = \frac{14}{243}$.

24. A certain family has 6 children, consisting of 3 boys and 3 girls. Assuming that all birth orders are equally likely, what is the probability the 3 eldest children are the 3 girls.

There are $\binom{6}{3}$ birth orders, as we choose the positions that the boys take up. Only one ordering has the 3 girls as the eldest, meaning the answer is $\frac{1}{20}$.

25. A city with 6 districts has 6 robberies in a particular week. Assume the robberies are located randomly, with all possibilities for which robberies occurred where equally likely. What is the probability that some district has more than one robbery?

If some district has more than one robbery, then it means it is not the case that each district has exactly one. So we can calculate p , the probability that there is a robbery in each district, and our answer will be $1 - p$.

The number of ways the robberies can place out is 6^6 , since the problem seems to imply that the robberies are numbered. Then, the number of sequences of robberies such that they happen in different districts is $6!$. So, $p = \frac{6!}{6^6} = \frac{5 \cdot 4}{6^4} = \frac{5}{324}$.

$1 - p$ gives $\frac{319}{324}$.

26. A survey is being conducted in a city with 1 million residents. It would be far too expensive to survey all the residents, so a random sample of 1000 is chosen (in practice there are many challenges with

sampling, such as obtaining a complete list of everyone in the city, and dealing with people who refuse to participate). The survey is conducted by choosing people one at a time, *with* replacement and equal probabilities.

- (a) Explain how sampling with vs. without replacement here relates to the birthday problem.
When sampling with replacement, this is similar to the birthday problem. Now, we have a room of 1000 people, and they can have one of 10^6 birthdays. If we sample without replacement, then it is not like the birthday problem, since it would mean everyone in the room would have different birthdays.
- (b) Find the probability that at least one person will get chosen more than once.

This is simply $1 - p$, where p is the probability that every person chosen is unique. In total, there are $(10^6)^{1000}$ ways to choose the people. However, for everyone to be unique, there are $10^6 \cdot (10^6 - 1) \cdot (10^6 - 2) \cdot \dots \cdot (10^6 - 999)$.

$$\text{This gives } 1 - p = \frac{(10^6)^{1000} - 10^6 \cdot (10^6 - 1) \cdot (10^6 - 2) \cdot \dots \cdot (10^6 - 999)}{(10^6)^{1000}}.$$

27. A *hash table* is a commonly used data structure in Computer Science, allowing for fast information retrieval. For example, suppose we want to store some people's phone numbers. Assume that no two of the people have the same name. For each name x , a *hash function* h is used, letting $h(x)$ be the location that will be used to store x 's phone number. After such a table has been computed, to look up x 's phone number one just recomputes $h(x)$ and then looks up what's stored in that location.

The hash function h is deterministic, since we don't want to get different results every time we compute $h(x)$. But h is often chosen to be *pseudorandom*. For this problem, assume true randomness is used. Let there be k people, with each person's phone number stored in a random location (with equal probabilities for each location, independently of where the other people's numbers are stored), represented by an integer between 1 and n . Find the probability that at least one location has more than one phone number stored there.

We compute p , the probability that all k hashes are unique. Our answer is $1 - p$.

Since all n hash locations are equiprobable, there are a total of n^k different hash functions. The probability that each person hashes to a unique location is then the number of ways to choose k from n and then permute them, which is $\binom{n}{k} k!$. Thus, our answer is $1 - \frac{\frac{n!}{(n-k)!}}{n^k} = 1 - \frac{n!}{n^k (n-k)!}$. Of course, if $k > n$, then a collision is guaranteed.

28. A college has 10 time slots for its courses, and blithely assigns courses to completely random time slots, independently. The college offers exactly 3 statistics courses. What is the probability that 2 or more of the statistics courses are in the same time slot?

If we compute p , the probability that all 3 statistics courses are assigned to unique timeslots, then our answer is $1 - p$.

Assuming the 3 statistics courses are distinguishable, then there are 10^3 ways to assign the statistics courses to timeslots. The number of ways such that they are all unique is $3! \binom{10}{3} = 720$. Thus, $p = \frac{720}{1000} = \frac{18}{25}$, so $1 - p = \frac{7}{25}$.

29. For each part, decide whether the blank should be filled in with $=$, $<$, or $>$, and give a clear explanation.

(a) (probability that the total after rolling 4 fair dice is 21) (*probability that the total after rolling 4 fair dice is 22*)

Here, both sides of the comparison occur in the same sample space of rolling 4 dice. We only need count whether there are more ways to make 21 vs 22.

We will casework based upon the number of 6's rolled. We start with rolling a sum of 21.

If 0 sixes are rolled, then it is impossible to reach a sum of 21, as the largest possible sum is $4 \cdot 5 = 20$.

If 1 six is rolled, then the remaining dice have to each roll 5's. If they roll anything less, then the total sum will not be 21. Thus, there are $\binom{4}{1} = 4$ ways to roll 21 with 1 six.

For 2 sixes, then the remaining two numbers must be 5 and 4 to make 9. Thus, the number of ways is $2\binom{4}{2} = 12$.

For 3 sixes, the last number must be a 3. Thus, there are $\binom{4}{3}$ ways.

Finally, for 4 sixes, the sum is 24, which is not 21.

Thus, we see there are 20 ways to roll 21 with 4 dice.

We repeat the process for 22.

For 0 sixes, it is impossible, and also for 1 six, as the largest sum possible in that case is 21.

For 2 sixes, the only way is for the other two dice to roll a 5, as anything smaller is less than 22. Thus, this gives $\binom{4}{2} = 6$ ways.

For 3 sixes, the last number must be a 4. So, there are $\binom{4}{3} = 4$ ways.

Once again, the sum of 4 sixes is 24, which is too big.

So, we see that for a sum of 22, there are 10 ways to do so using 4 dice. Thus, because $10 < 20$, the probability of rolling a sum of 21 is higher than the probability of rolling a sum of 22 using 4 dice. The answer is $>$.

(b) (probability that a random 2-letter word is a palindrome³) (*probability that a random 3-letter word is not a palindrome*)

³ A *palindrome* is an expression such as "A man, a plan, a canal: Panama" that reads the same backwards as forwards (ignoring spaces, capitalization, and punctuation). Assume for this question that all words of the specified length are equally likely, that there are no spaces or punctuation, and that the alphabet consists of lowercase letters $a, b, \dots, z$. A word is any string of letters from the alphabet; it does not need to be a word that has meaning in the English language.

We first compute the probability that a random 2-letter word is a palindrome:

In this case the two letters must be the same, so there are 26 palindromes over 26^2 possibilities. So, the answer is $\frac{1}{26}$.

The probability that a 3-letter word is not a palindrome is $1 - p$, where p is the probability that a 3-letter word is a palindrome. For a 3-letter word to be a palindrome, all that is required is for the first and last

characters to be the same. This gives 26^2 palindromes, as we multiply the number of ways to choose the first/third letter by the number of ways to choose the middle one. There are 26^3 3-letter words, meaning our answer is $\frac{1}{26}$ again. Thus, the answer is $=$.

30. With definitions as in the previous problem, find the probability that a random n -letter word is a palindrome for $n = 7$ and $n = 8$.

For $n = 7$, there are 26^7 7-letter words and 26^4 palindromes, since we choose the 1st and 7th letter, the 2nd and 6th letter, the 3rd and 5th letter, and the 4th letter. Our answer is $\frac{1}{26^3}$.

For $n = 8$, there are 26^8 8-letter words and 26^4 palindromes still (since we still only make 4 choices for the 1st and 8th, 2nd and 7th, 3rd and 6th, and 4th and 5th). So, our answer is $\frac{1}{26^4}$.

31. Elk dwell in a certain forest. There are N elk, of which a simple random sample of size n are captured and tagged ("simple random sample" means that all $\binom{N}{n}$ sets of n elk are equally likely). The captured elk are returned to the population, and then a new sample is drawn, this time with size m . This is an important method that is widely used in ecology, known as *capture-recapture*. What is the probability that exactly k of the m elk in the new sample were previously tagged? (Assume that an elk that was captured before doesn't become more or less likely to be captured again).

There are n elk in the forest which have been tagged. Since all $\binom{N}{m}$ samples are equiprobable, we want to know how many ways there are to choose exactly k of the n tagged elk. This is of course $\binom{n}{k} \cdot \binom{N-n}{m-k}$, so our answer is $\frac{\binom{n}{k} \cdot \binom{N-n}{m-k}}{\binom{N}{m}}$.

32. Four cards are face-down on the table. You are told that two are red and two are black, and you need to guess which two are red and which two are black. You do this by pointing to the two cards you're guessing are red (and then implicitly you're guessing that the other two are black). Assume that all configurations are equally likely, and that you do not have psychic powers. Find the probability that exactly j of your guesses are correct, for $j = 0, 1, 2, 3, 4$.

Let's first compute the probability $j = 0$. In such a case, we guess the exact opposite of the actual configuration, and there is only one way to do this. Since there are $\binom{6}{4}$ choices we could make, it's a $\frac{1}{6}$ chance to guess 0 correct. Similarly, we have a $\frac{1}{6}$ chance to guess all 4 correct.

It is not possible to guess 3 right, as if 3 are right, then we must have also guessed the fourth right. Similarly, if we guess 1 correct, meaning 3 wrong, then the last one must also be wrong, as there is only one option left for what we can guess and what it could be, and they differ. Thus, the only remaining j is 2, so $j = 2$ occurs with probability $1 - \frac{1}{6} - \frac{1}{6} = \frac{2}{3}$.

33. A jar contains r red balls and g green balls, where r and g are fixed positive integer. A ball is drawn from the jar randomly (with all possibilities equally likely), and a second ball is drawn randomly.

- (a) Explain intuitively why the probability of the second ball being green is the same as the probability of the first ball being green.

Let's arbitrarily label the balls.

Picking two balls without replacement is just choosing 2 from $r + g$, but we order the two balls randomly. Of course, for any combination of red and green, the probability that ball 1 is green is the same as ball 2 being green, so the probability in general that ball 1 is green is the same as the probability that ball 2 is green.

- (b) Define notation for the sample space of the problem, and use this to compute the probabilities from (a) and show that they are the same.

Let the ball selections be described as a tuple (a, b) , where a is the color of ball 1 and b the color of ball 2. Then, the sample space is $(r, r), (r, g), (g, r), (g, g)$. The probability that ball 1 is green is $P(g, r) + P(g, g)$, while the probability that ball 2 is green is $P(r, g) + P(g, g)$.

We can show that $P(g, r) = P(r, g)$: There are in total $(r + g)(r + g - 1)$ ways to select balls for an arbitrarily labeled set of r red and g green balls, and the number of ways such that the first is green and second is red is rg , as is the number of ways the first is red and the second is green. As such, both probabilities are the same.

Since we have $P(g, r) = P(r, g)$, we have $P(g, r) + P(g, g) = P(r, g) + P(g, g)$.

- (c) Suppose there are 16 balls in total, and that the probability that the two balls are the same color is the same as the probability that they are different colors. What are r and g (list all possibilities)?

Using notation from the previous part, we have that $P(r, g) + P(g, r) = P(r, r) + P(g, g)$. Since $P(r, r) + P(g, g) + P(r, g) + P(g, r) = 1$, $P(r, r) + P(g, g) = \frac{1}{2}$.

We have

$$\begin{aligned}\frac{r(r-1)}{(r+g)(r+g-1)} + \frac{g(g-1)}{(r+g)(r+g-1)} &= \frac{1}{2} \\ 2r(r-1) + 2g(g-1) &= (r+g)(r+g-1) \\ 2r^2 - 2r + 2g^2 - 2g &= r^2 + gr - r + gr + g^2 - g \\ r^2 - r + g^2 - g &= 2gr\end{aligned}$$

Recall $r + g = 16$, so $r = 16 - g$. Thus,

$$\begin{aligned}(16 - g)^2 - (16 - g) + g^2 - g &= 2g(16 - g) \\ 256 - 32g + g^2 - 16 + g + g^2 - g &= 32g - 2g^2 \\ 4g^2 - 64g + 240 &= 0 \\ g^2 - 16g + 60 &= 0 \\ (g - 6)(g - 10) &= 0\end{aligned}$$

Thus, we could have $g = 6, r = 10$ or $r = 6, g = 10$, which we can verify since $\frac{10 \cdot 9}{16 \cdot 15} + \frac{6 \cdot 5}{16 \cdot 15} = \frac{120}{240} = \frac{1}{2}$.

34. A random 5-card poker hand is dealt from a standard deck of cards. Find the probability of each of the following possibilities (in terms of binomial coefficients)

- (a) A flush (all 5 cards being the same suit; do not count a royal flush, which is a flush with an ace, king, queen, jack, 10).

If all hands are equiprobable and the total number of hands is $\binom{52}{5}$, then the number of flushes is $4 \cdot \binom{13}{5}$, since there are 4 suits and $\binom{13}{5}$ flushes per suit. However, we must subtract the 4 royal flushes of each suit, giving $4(\binom{13}{5} - 1)$ valid flushes. Our answer is then $\frac{4(\binom{13}{5} - 1)}{\binom{52}{5}}$.

- (b) Two pair (e.g., two 3's, two 7's, and an ace).

To count the number of two pair hands, we first count the number of ways to choose the "pair" numbers and the extra. This is $\binom{13}{3}$ multiplied by 3, since we choose one to be the extra. Then, the number of hands that satisfy this 'pair' and extra is $\binom{4}{2} \cdot \binom{4}{2} \cdot 4 = 144$, since we choose which two suits of the pairs we want and choose the suit for the extra.

Thus, our answer is $\frac{432\binom{13}{3}}{\binom{52}{5}}$.

35. A random 13-card hand is dealt from a standard deck of cards. What is the probability that the hand contains at least 3 cards of every suit?

Since each of the 4 suits has at least 3 cards, they will take up 12 of our 13 cards. This indicates that to have at least 3 cards per suit, we will have 3 suits with 3 cards and one suit with 4. There are 4 ways to choose the suit which has extra. Then, for the suits with 3 cards, there are $\binom{13}{3}$ ways to choose the cards, and $\binom{13}{4}$ ways to choose the suit with extra cards.

Thus, there are $4 \cdot \binom{13}{3}^3 \binom{13}{4}$ hands, out of $\binom{52}{13}$. Thus, our answer is $\frac{4 \cdot \binom{13}{3}^3 \cdot \binom{13}{4}}{\binom{52}{13}}$.

36. A group of 30 dice are thrown. What is the probability that 5 of each of the values 1, 2, 3, 4, 5, 6 appear?

There are 6^{30} different outcomes in total. The number of ways to have 5 each of everything is $\binom{30}{5} \cdot \binom{25}{5} \cdot \binom{20}{5} \cdot \binom{15}{5} \cdot \binom{10}{5}$, so our answer is $\frac{\binom{30}{5} \cdot \binom{25}{5} \cdot \binom{20}{5} \cdot \binom{15}{5} \cdot \binom{10}{5}}{6^{30}} = \frac{\frac{30!}{(5!)^6}}{6^{30}} = \frac{30!}{(5!)^6 \cdot 6^{30}}$.

37. A deck of cards is shuffled well. The cards are dealt one by one, until the first time an ace appears.

- (a) Find the probability that no kings, queens, or jacks appear before the first ace.

There are $52!$ total shufflings possible. We need to count how many permutations have the first ace before the first queen, king, jack.

We first choose 16 positions to contain the 4 aces, queens, kings, and jack: $\binom{52}{16}$. We then permute the other 36 cards, giving $\frac{52!}{16!}$.

We now count the number of ways to order just the aces, kings, queens, and jacks. There are 4 ways to choose the first ace, then $15!$ ways to order the remaining cards, so the number of shufflings that have the first ace before the first king, queen, and jack is $4 \cdot 15! \cdot \frac{52!}{16!}$.

Our answer is $\frac{4 \cdot 15! \cdot 52!}{16! \cdot 52!} = \frac{4 \cdot 15!}{16!} = \frac{1}{4}$.

- (b) Find the probability that exactly one king, exactly one queen, and exactly one jack appear (in any order) before the first ace.

There are a total of $52!$ ways the deck could have been shuffled. We must count the number of orderings such that there is exactly one of each of the face cards before the first ace.

We first choose the positions which the face cards + aces take up: $\binom{52}{16}$. We then permute the non-face cards and aces, giving $\binom{52}{16} \cdot (36)! = \frac{52!}{16!}$. Then, there are $(4!)^4$ ways to choose the relative ordering of each number. For example, $4!$ ways to order the kings, $4!$ ways to order the queens, etc.

Positions 1, 2, 3 are going to be the first of the aces, kings, and queens, so there are $3!$ ways to choose the first 4 positions (the 4th will be the first ace). Then, For the remaining 12 positions, we choose 3 to be kings, 3 to be queens, and 3 to be jacks, as we have already ordered them. This gives $3! \cdot \binom{12}{3} \cdot \binom{9}{3} \cdot \binom{6}{3} = 3! \cdot \frac{12!}{(3!)^4} = \frac{12!}{216}$.

So, there are in total $\frac{52!}{16!} \cdot (4!)^4 \cdot \frac{12!}{216}$ ways to have one king, one queen, and one jack appear before the first ace.

Our final answer is $\frac{52! \cdot (4!)^4 \cdot 12!}{16! \cdot 216 \cdot 52!} = \frac{(4!)^4 12!}{216 \cdot 16!} = \frac{16}{455}$.

38. Tyrion, Cersei, and ten other people are assigned a round table, with their seating arrangement having been randomly assigned. What is the probability that Tyrion and Cersei are sitting next to each other? Find this in two ways:

- (a) using a sample space of size $12!$, where an outcome is fully detailed about the seating;

We count the number of permutations of 12 people where Tyrion and Cersei are next to each other. There are 12 ways to choose the two seats they will sit in and 2 ways to order them in those seats. Finally, there are $10!$ ways to order the remaining people.

This gives us $\frac{12 \cdot 2 \cdot 10!}{12!} = \frac{2}{11}$.

- (b) using a much smaller sample space, focusing on Tyrion and Cersei.

Assume Tyrion and Cersei are the first to sit down. Tyrion chooses any seat. Then, Cersei has 11 choices, all equiprobably. 2 of those choices leave her next to Tyrion, so the answer is $\frac{2}{11}$.

39. An organization with $2n$ people consists of n married couples. A committee of size k is selected, with all possibilities equally likely. Find the probability that there are exactly j married couples within the committee.

The committee must have j married couples, of which there are $\binom{n}{j}$ ways to select them. From the remaining $n - j$ couples, we must select at most one from each of the remaining couples till we have k people. Thus, we must select $k - 2j$ from $n - j$. However, after selecting which couples to choose one person from, there are 2^{k-2j} ways to choose them.

So, there are $\binom{n}{j} \cdot \binom{n-j}{k-2j} \cdot 2^{k-2j}$ committees which have j married couples. There are $\binom{2n}{k}$ ways to choose a committee, so our answer is $\frac{\binom{n}{j} \cdot \binom{n-j}{k-2j} \cdot 2^{k-2j}}{\binom{2n}{k}}$.

40. There are n balls in a jar, labeled with the numbers $1, 2, \dots, n$. A total of k balls are drawn, one by one *with replacement*, to obtain a sequence of numbers.

(a) What is the probability the sequence obtained is strictly increasing?

There are n^k possible sequences. The number of strictly increasing sequences of length k is $\binom{n}{k}$. So, our answer is $\frac{\binom{n}{k}}{n^k}$.

(b) What is the probability that the sequence obtained is increasing? (but not necessarily strictly increasing, i.e., there can be repetitions)?

The number of possible sequences is still n^k . However, if the sequences do not need to be strictly increasing, we instead have one sequence for every nonnegative integer solution to $x_1 + x_2 + \dots + x_n = k$, which is $\binom{n+k-1}{k}$. Thus, our answer is $\frac{\binom{n+k-1}{k}}{n^k}$.

41. Each of n balls is independently placed into one of n boxes, with all boxes equally likely. What is the probability that exactly one box is empty?

The number of ways to put n balls in n boxes is n^n if the balls are labeled (only in this scenario can all outcomes be equiprobable).

Then, for the number of ways to place the balls such that exactly one box is empty, we must first choose which box is empty, where there are n ways to do this. Since we have n balls for $n - 1$ remaining boxes, and none of these $n - 1$ boxes can be empty, there is exactly one box which will have 2 balls. There are $n - 1$ ways to choose this box.

We must choose two of the n balls to put in the extra box, which is $\binom{n}{2}$, and then we have $(n - 2)!$ ways to assign the remaining balls to boxes.

The number of ways is thus $n(n - 1) \cdot \binom{n}{2} \cdot (n - 2)! = n(n - 1) \cdot \frac{n!}{2!(n - 2)!} \cdot (n - 2)! = \frac{n(n - 1)n!}{2}$. So, our answer is $\frac{\frac{n(n - 1)n!}{2}}{n^n} = \frac{n(n - 1)n!}{2n^n} = \frac{(n - 1)n!}{2n^{n - 1}}$.

42. A *norepeatword* is a sequence of at least one (and possibly all) of the usual 26 letters $a, b, c, \dots, z$ with repetitions not allowed. For example, "course" is a norepeatword, but "statistics" is not. Order matters, e.g., "course" is not the same as "source". A norepeatword is chosen randomly, with all norepeatwords equally likely. Show the probability that it uses all 26 letters is very close to $\frac{1}{e}$.

If the norepeatword is length k , then there are $\binom{26}{k}k! = \frac{26!}{(26 - k)!}$ norepeatwords that long. Thus, the number of norepeatwords is

$$\sum_{k=1}^{26} \frac{26!}{(26 - k)!}$$

The number of norepeatwords of length 26 is $26!$. Thus, the probability we are interested in is

$$\frac{26!}{\sum_{k=1}^{26} \frac{26!}{(26 - k)!}}$$

$$\frac{26!}{26! \sum_{i=1}^{26} \frac{1}{(26-i)!}}$$

$$\frac{1}{\sum_{i=1}^{26} \frac{1}{(26-i)!}}$$

$$\frac{1}{\sum_{i=0}^{25} \frac{1}{i!}}$$

Note that $\sum_{i=0}^{25} \frac{1}{i!}$ is an approximation for e^1 since it is the first 26 terms to the infinite Taylor series of e^x for $x = 1$ (which is $\sum_{i=0}^{\infty} \frac{x^i}{i!}$). Thus, $\sum_{i=0}^{25} \frac{1}{i!} \approx e$ and $\frac{1}{\sum_{i=0}^{25} \frac{1}{i!}} \approx \frac{1}{e}$.

4 Axioms of probability

43. Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B)$$

We know $P(A) + P(B) - P(A \cap B) = P(A \cup B)$. Since $P(A \cup B)$ is a probability, $P(A \cup B) \leq 1$, so $P(A) + P(B) - P(A \cap B) \leq 1$, meaning $P(A) + P(B) - 1 \leq P(A \cap B)$. If $P(A \cup B) = 1$, then $P(A) + P(B) - 1 = P(A \cap B)$, so $P(A \cup B) = 1$ is the condition for equality.

$P(A \cap B) \leq P(A \cup B)$ is easy to see since $A \cap B \subset A \cup B$, so $P(A \cap B) \leq P(A \cup B)$. They are equal when $A \cap B = A \cup B$, which would mean $A = B$. If $A \neq B$, then there is some element x such that $x \in A$ and $x \notin B$ or $x \notin A$ and $x \in B$, so $A \cap B \neq A \cup B$.

$P(A \cup B) \leq P(A) + P(B)$. Since $P(A) + P(B) - P(A \cap B) = P(A \cup B)$, and $P(A \cap B) \geq 0$, it must be the case that $P(A) + P(B) \geq P(A \cup B)$. $P(A) + P(B) = P(A \cup B)$ when $P(A \cap B) = 0$.

44. Let A and B be events. The *difference* $B - A$ is defined to be the set of all elements of B that are not in A . Show that if $A \subset B$, then

$$P(B - A) = P(B) - P(A)$$

directly using the axioms of probability.

We know for disjoint sets X, Y $P(X) + P(Y) = P(X \cup Y)$. Here, let $X = A$ and $Y = B - A$. They are indeed disjoint, as if $x \in A$, then $x \notin B - A$ and if $x \in B - A$, then $x \notin A$.

Thus, we have $P(A) + P(B - A) = P(A \cup B) = P(B)$, giving $P(B) - P(A) = P(B - A)$.

45. Let A and B be events. The *symmetric difference* $A \triangle B$ is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the XOR (*exclusive or*) of A and B . Show that

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B)$$

directly using the axioms of probability.

Since $A \Delta B$ and $A \cap B$ are disjoint, $P(A \Delta B) + P(A \cap B) = P((A \Delta B) \cup (A \cap B)) = P(A \cup B)$. Furthermore, we know that $P(A) + P(B) - P(A \cap B) = P(A \cup B)$, so

$$P(A) + P(B) - P(A \cap B) - P(A \cap B) = P(A \cup B) - P(A \cap B)$$

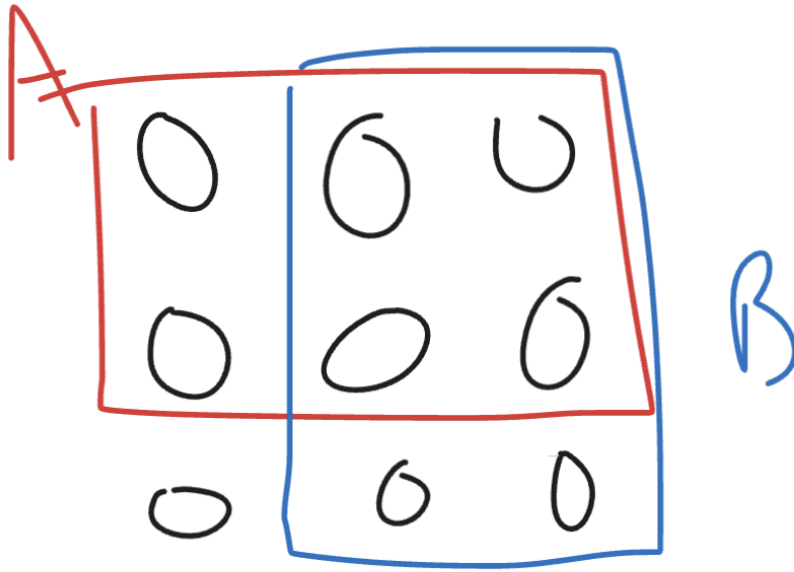
$$P(A) + P(B) - 2P(A \cap B) = P(A \Delta B)$$

46. Let $A_1, A_2, \dots, A_n$ be events. Let B_k be the event exactly k of A_i occur, and C_k be the event that at least k of the A_i occur, for $0 \leq k \leq n$. Find a simple expression for B_k in terms of $P(C_k)$ and $P(C_{k+1})$.

We have already proved in question 44 that if $A \subset B$, then $P(B - A) = P(B) - P(A)$. We can note that $C_{k+1} \subset C_k$, and $C_k - C_{k+1} = B_k$. Thus, $P(B_k) = P(C_k) - P(C_{k+1})$.

47. Events A and B are independent if $P(A \cap B) = P(A)P(B)$ (independence is explored in detail next chapter).

- (a) Give an example of independent events A and B in a finite sample space S (with neither equal to $\emptyset$ or S), and illustrate it with a Pebble World diagram.



- (b) Consider the experiment of picking a point randomly in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

When the area of the intersection of A_1 and B_1 is equal to the area of A_1 times the area of B_1 , then A and B are independent.

If A_1 were the rectangle with bottom left corner $(0,0)$ and top right corner $(1, \frac{1}{2})$ and B_1 were the rectangle with bottom left corner $(0,0)$ and top right corner $(\frac{1}{2}, 1)$, then the area of A_1 and B_1 would be $\frac{1}{2}$, while the area of their intersection is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

A scenario where A and B are not independent is if we keep the same A from the previous situation but let B be the rectangle with bottom left corner $(0, \frac{1}{2})$ and top right corner $(1, 1)$.

(c) Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c)$$

Since $P(A \cap B) = P(A)P(B)$ for A, B independent, $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$. Since $P(X) = 1 - P(X^c)$, we have

$$P(A) + P(B) - P(A) \cdot P(B) = (1 - P(A^c)) + (1 - P(B^c)) - (1 - P(A^c))(1 - P(B^c))$$

$$P(A) + P(B) - P(A)P(B) = 2 - P(A^c) - P(B^c) - (1 - P(A^c) - P(B^c) + P(A^c)P(B^c))$$

$$P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c)$$

48. Arby has a belief system assigning a number $P_{Arby}(A)$ between 0 and 1 to every event A (for some sample space). This represents Arby's degree of belief about how likely A is to occur. For any event A , Arby is willing to pay a price of $1000 \cdot P_{Arby}(A)$ dollars to buy a certificate as shown be-

Certificate

The owner of this certificate can redeem it for \$1000 if A occurs. No value if A does not occur, except as required by federal, state, or local law. No expiration date.

low:

Likewise, Arby is willing to sell such a certificate at the same price. Indeed, Arby is willing to buy and sell any number of certificates at this price, as Arby considers it the "fair" price.

Arby stubbornly refuses to accept the axioms of probability. In particular, suppose there are two *disjoint* events A and B with

$$P_{Arby}(A \cup B) \neq P_{Arby}(A) + P_{Arby}(B)$$

Show how to make Arby go bankrupt, by giving a list of transactions Arby is willing to make that will *guarantee* that Arby will lose money (you can assume it will be known whether A occurred and whether B occurred the day after any certificates are bought/sold).

There are two cases, depending on whether $P(A) + P(B) + P((A \cup B)^c)$ is greater than, less than, or equal to 1.

If $P(A) + P(B) + P((A \cup B)^c) < 1$, then we can buy all three of $A, B, (A \cup B)^c$ from Arby and he will lose money, as the next day we will earn 1000 dollars from him, since $P(A \cup B \cup (A \cup B)^c) = 1$.

If $P(A) + P(B) + P((A \cup B)^c) > 1$, then we can sell all three to Arby, after which he will gain exactly 1000 the next day, not enough to make up for the deficit.

If $P(A) + P(B) + P((A \cup B)^c) = 1$, then because $P(A \cup B) \neq P(A) + P(B)$, we have $P(A \cup B) + P((A \cup B)^c) \neq 1$. We follow the same logic as before, selling if the sum is greater than 1 and buying if the sum is less than 1.

5 Inclusion-Exclusion

49. A fair die is rolled n times. What is the probability that at least 1 of the 6 values never appears?

We will use the principle of inclusion exclusion. Let $P(S)$ be the probability that no values in S appear. Our answer is

$$\begin{aligned} & \sum_{S \subset \{1,2,3,4,5,6\}} (-1)^{|S|+1} \cdot P(S) \\ & \sum_{S \subset \{1,2,3,4,5,6\}} (-1)^{|S|+1} \cdot \frac{(6-|S|)^n}{6^n} \\ & \frac{1}{6^n} \cdot \sum_{|S|=1}^6 (-1)^{|S|+1} \cdot \binom{6}{|S|} \cdot (6-|S|)^n \\ & \frac{1}{6^n} \cdot \sum_{|S|=1}^6 (-1)^{|S|+1} \cdot \binom{6}{6-|S|} \cdot (6-|S|)^n \\ & \frac{1}{6^n} \cdot \sum_{k=0}^5 (-1)^{k+1} \cdot \binom{6}{k} \cdot k^n \end{aligned}$$

Our answer is $\frac{0^n+6-15 \cdot 2^n+20 \cdot 3^n-15 \cdot 4^n+6 \cdot 5^n}{6^n}$.

50. A card player is dealt a 13-hand from a well-shuffled, standard deck of cards. What is the probability the hand is void in at least one suit ("void in a suit" means having no cards of that suit)?

If $P(S)$ is the probability that no cards of any suit in S is present in the hand. Using the principle of inclusion exclusion gives

$$\begin{aligned} & \sum_{S \subset \{\text{Heart}, \text{Spade}, \text{Diamond}, \text{Club}\}} (-1)^{|S|+1} \cdot P(S) \\ & \sum_{S \subset \{\text{Heart}, \text{Spade}, \text{Diamond}, \text{Club}\}} (-1)^{|S|+1} \cdot \frac{\binom{52-13|S|}{13}}{\binom{52}{13}} \\ & \sum_{|S|=1}^4 (-1)^{|S|+1} \cdot \frac{\binom{52-13|S|}{13}}{\binom{52}{13}} \cdot \binom{4}{|S|} \end{aligned}$$

Thus, our answer is $\frac{4\binom{39}{13}+6\binom{26}{13}+4\binom{13}{13}}{\binom{52}{13}}$.

51. For a group of 7 people, find the probability that all 4 seasons (winter, spring, summer, fall) occur at least once each among their birthdays, assuming that all seasons are equally likely.

The answer is $1 - p$, where p is the probability that at least one season does not appear amongst their birthdays.

We can compute p using the principle of inclusion-exclusion.

$$\sum_{S \subset \{Spring, Fall, Winter, Summer\}} (-1)^{|S|+1} \frac{(4 - |S|)^7}{4^7}$$

$$\frac{1}{4^7} \cdot \sum_{|S|=1}^4 (-1)^{|S|+1} (4 - |S|)^7 \cdot \binom{4}{|S|}$$

$$\frac{1}{4^7} \sum_{k=0}^3 (-1)^{k+1} k^7 \cdot \binom{4}{k}$$

Thus, $p = \frac{4-2^7 \cdot 6+3^7 \cdot 4}{4^7}$, so $1 - p = \frac{4^7 - (4-2^7 \cdot 6+3^7 \cdot 4)}{4^7} = \frac{525}{1024}$.

52. A certain class has 20 students, and meets on Mondays and Wednesdays in a classroom with exactly 20 seats. In a certain week, everyone in the class attends both days. On both days, the students choose their seats completely randomly (with one student per seat). Find the probability that no one sits in the same seat on both days of that week.

Using the Principle of Inclusion-Exclusion, the answer is

$$\sum_{S \subset \{1, 2, \dots, 20\}} (-1)^{|S|} \cdot \frac{(20 - |S|)!}{20!}$$

Here, for a subset of size S , we are counting the number of ways all students in $|S|$ sit in the same seat both days. Based on where they sit on Monday, we can permute the students not in S $(20 - |S|)!$ ways, as the students in $|S|$ should sit in the same seat. Of course, there are $20!$ ways to permute the children on Wednesday in general.

$$\sum_{|S|=0}^{20} (-1)^{|S|} \cdot \frac{(20 - |S|)!}{20!} \cdot \binom{20}{|S|}$$

$$\frac{1}{(20!)^2} \cdot \sum_{k=0}^{20} (-1)^k \cdot \frac{1}{k!}$$

So, our answer is $\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{1}{18} - \frac{1}{19!} + \frac{1}{20!}$.

53. Fred needs to choose a password for a certain website. Assume he will choose an 8-character password, and that the legal characters are the lowercase letters $a, b, c, \dots, z$, the uppercase letters $A, B, C, \dots, Z$ and the numbers $0, 1, \dots, 9$.

- (a) How many possibilities are there if he is required to have at least one lowercase letter in his password?

There are $26 + 26 + 10 = 62$ choices for each character. If the password has 8 letters, there are 62^8 passwords. However, we want to know how many have at least one lowercase letter, which is 62^8 minus the number of passwords with no lowercase letters, giving $62^8 - 36^8$.

- (b) How many possibilities are there if he is required to have at least one lowercase letter and at least one uppercase letter in his password?

We still have 62^8 minus the number of passwords that do not have both at least one lowercase and uppercase letter. We subtract 36^8 twice since 36^8 is both the number of passwords with no

lowercase letters and the number of passwords with no uppercase letters. However, we subtracted the passwords that have neither twice, so we must add back 10^8 , the number of passwords that are only digits.

Thus, the answer is $62^8 - 2 \cdot 36^8 + 10^8$.

- (c) How many possibilities are there if he is required to have at least one lowercase letter, at least one uppercase letter, and at least one number in his password?

We use Principle of Inclusion-Exclusion to count the union of no lowercase letter, no uppercase letter, and no number.

We first have the size of each individual set (no lowercase, no uppercase, no number) $36^8 + 36^8 + 52^8$, then subtract every combination of two which is (no uppercase and letter, no lowercase and letter, no letters) $-26^8 - 26^8 - 10^8$ and then finally add back the number of passwords that don't contain anything, which is 0. So, the number of passwords which contain no lowercase, no uppercase, or no number is $2 \cdot 36^8 + 52^8 - 2 \cdot 26^8 - 10^8$.

Subtracting this from 62^8 , the total number of passwords, we get the number of passwords that aren't missing uppercase, lowercase, and numbers, meaning they have at least one of each: $62^8 - 2 \cdot 36^8 - 52^8 + 2 \cdot 26^8 + 10^8$.

54. Alice attends a small college in which each class meets only once a week. She is deciding between 30 non-overlapping classes. There are 6 classes to choose from for each day of the week, Monday through Friday. Trusting in the benevolence of randomness, Alice decides to register for 7 randomly selected classes out of the 30, with all choices equally likely. What is the probability she will have classes every day, Monday through Friday? (This problem can be done either directly using the naive definition of probability, or using inclusion-exclusion).

Since there are 7 classes, and 5 of them must be from Monday through Friday, so the other two extra classes have two cases. Either there are two days with two classes, or one day with three classes. We count these cases separately.

If two days have two classes, we have $\binom{5}{2}$ ways to choose which 2 days get extra, and then we have $6^3 \cdot \binom{6}{2}^2$ ways to choose the courses. We have 6 choices for the days which have one class, and $\binom{6}{2}$ choices for the days which have 2. So, the total number is $\binom{5}{2} \cdot \binom{6}{2}^2 \cdot 6^3$.

If one day has three classes, there are 5 days to choose from, and then for the other 4 days 6^4 ways to choose their class and then $\binom{6}{3}$ selections for the 3 courses on the extra day. This gives $5 \cdot 6^4 \cdot \binom{6}{3}$.

So, we have $\binom{5}{2} \cdot \binom{6}{2}^2 \cdot 6^3 + 5 \cdot 6^4 \cdot \binom{6}{3}$ total ways, and the probability is $\frac{\binom{5}{2} \cdot \binom{6}{2}^2 \cdot 6^3 + 5 \cdot 6^4 \cdot \binom{6}{3}}{\binom{30}{7}} = \frac{114}{377}$.

55. A club consists of 10 seniors, 12 juniors, and 15 sophomores. An organizing committee of size 5 is chosen randomly (with all subsets of size 5 equally likely).

- (a) Find the probability that there are exactly 3 sophomores in the committee.

There are $\binom{37}{5}$ ways to choose a committee, but for 3 sophomores there are $\binom{15}{3} \cdot \binom{22}{2}$ ways to choose a committee with 3 sophomores.

So, the answer is $\frac{\binom{15}{3} \cdot \binom{22}{2}}{\binom{37}{5}}$.

- (b) Find the probability that the committee has at least one representative from each of the senior, junior, and sophomore classes.

We use the principle of inclusion exclusion to take the complement of the union of the probability that the senior, junior, and sophomore classes are not present. Let $P(S)$ be the probability that none of the classes in S are present in the committee.

There are a total of $\binom{37}{5}$ committees. We subtract $\binom{22}{5}$ no-sophomore committees, $\binom{25}{5}$ no-junior committees, and $\binom{27}{5}$ no-senior committees. However, we must then add back senior only, junior only, and sophomore only committees that have been subtracted twice.

So, our answer is $\binom{37}{5} - \binom{22}{5} - \binom{25}{5} - \binom{27}{5} + \binom{10}{5} + \binom{12}{5} + \binom{15}{5}$. We divide by the number of total committees, giving $\frac{\binom{37}{5} - \binom{22}{5} - \binom{25}{5} - \binom{27}{5} + \binom{10}{5} + \binom{12}{5} + \binom{15}{5}}{\binom{37}{5}}$.

6 Mixed Practice

56. For each part, decide whether the blank should be filled in with $=$, $<$, or $>$, and give a clear explanation. In (a) and (b), order doesn't matter.

- (a) (number of ways to choose 5 people out of 10) *(number of ways to choose 6 people out of 10)*

The LHS is $\binom{10}{5} = \frac{10!}{5! \cdot 5!}$ and the RHS is $\binom{10}{6} = \frac{10!}{6! \cdot 4!}$. $6! \cdot 4! > 5! \cdot 5!$, since $5! \cdot 5! \cdot \frac{6}{5} = 6! \cdot 4!$. Since $\binom{10}{5}$ has the smaller denominator, it must be larger than $\binom{10}{6}$, so the answer is $>$.

- (b) (number of ways to break 10 people into 2 teams of 5) *(number of ways to break 10 people into a team of 6 and a team of 4)*

Since order doesn't matter, on the left is $\binom{10}{5}$ and on the right is $\binom{10}{6}$, and from the previous problem we know the left hand side is bigger, so the answer is $>$.

- (c) (probability that all 3 people in a group of 3 people were born on January 1) *(probability that in a group of 3 people, 1 was born on a day other than January 1)*

We imagine the birthdays of the people as a 3-tuple. For the LHS, there is only one valid tuple: (January 1, January 1, January 1), while on the RHS there are six valid tuples due to there being $3!$ ways to order the 3 birthdays. Thus, the answer is $<$.

- (d) Martin and Gale play an exciting game of "toss the coin", where they toss a fair coin until the pattern HH occurs (two consecutive heads) or the pattern TH occurs (Tails followed immediately by Heads). Martin wins the game if and only if HH occurs before TH occurs.

(probability that Martin wins) $1\frac{1}{2}$

If the first flip is a T , then Martin must win, since if any H is ever flipped, Martin will immediately win before Gale gets his HH .

If the first flip is an H , then if the next flip is an H Gale will win, while if the next flip is a T once again Martin wins for sure even if not immediately. Thus, Martin wins with probability greater than $\frac{1}{2}$.

57. Take a deep breath before attempting this problem. In the book *Innumeracy* [20], John Allen Paulos writes:

Now for better news a kind of immortal persistence. First, take a deep breath. Assume Shakespeare's account is accurate and Julius Caesar gasped ["Et tu, Brute!"] before breathing his last. What are the chances you just inhaled a molecule which Caesar exhaled in his dying breath?

Assume that one breath of air contains 10^{22} molecules and that there are 10^{44} molecules in the atmosphere. (These are slightly simpler numbers than the estimates that Paulos gives; for the purposes of the problem, assume these are exact. Of course, in reality there are many complications such as different types of molecules in the atmosphere, chemical reactions, variation in lung capacities, etc.)

Suppose that the molecules in the atmosphere now are the same as those in the atmosphere when Caesar was alive, and that in the 2000 years or so since Caesar, these molecules have been scattered completely randomly through the atmosphere. Also, assume that Caesar's last breath was sampled *without* replacement but that your breathing is sampled *with* replacement (without replacement makes more sense but with replacement is easier to work with, and is a good approximation since the number of molecules in the atmosphere is so much larger than the number of molecules in one breath).

Find the probability that at least one molecule in the breath you just took was shared with Caesar's last breath, and give a simple approximation in terms of e .

Hint: As discussed in the math appendix, $(1 + \frac{x}{n})^n \approx e^x$ for n large.

Of the 10^{44} molecules in the atmosphere, 10^{22} of them were in Caesar's last breath. We draw 10^{22} molecules with replacement from the atmosphere.

For each molecule, there is a $\frac{10^{22}}{10^{44}} = \frac{1}{10^{22}}$ chance it was in Caesar's last breath, so the probability it wasn't is $1 - \frac{1}{10^{22}}$. For none of the molecules to be from Caesar gives a probability of $(1 - \frac{1}{10^{22}})^{(10^{22})} \approx e^{-1}$, since $\lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n = e^x$. Thus, our answer is $1 - e^{-1} = 1 - \frac{1}{e}$.

58. A widget inspector inspects 12 widgets and finds that exactly 3 are defective. Unfortunately, the widgets get all mixed up and the inspector has to find the 3 defective widgets again by testing the widgets one by one.

- (a) Find the probability that the inspector will now have to test at least 9 widgets.

If we imagine that in the order of the widgets, at least one is in the last four, then we will have to test at least 9 widgets. This is $1 - p$, where p is the probability all three defective widgets are in the first 9.

Since there are $\binom{12}{3}$ ways to pick the defective widgets, and $\binom{8}{3}$ ways to put the defective widgets in the first 8 spots, the answer is $1 - \frac{\binom{8}{3}}{\binom{12}{3}} = \frac{41}{55}$.

- (b) Find the probability that the inspector will now have to test at least 10 widgets.

The inspector will need to test 10 widgets if he doesn't find the all defective widgets in the first nine he tests or if he finds no defective widgets in the first 9 he tests. Thus, we first take the probability that the defective widgets lie in the first 9, which is $\frac{\binom{9}{3}}{\binom{12}{3}}$, which would mean we do less than 10 inspections, and we add $\frac{1}{\binom{12}{3}}$, which is the probability that all 3 bad widgets are last. This gives $\frac{\binom{9}{3}+1}{\binom{12}{3}} = \frac{21}{55}$.

We subtract this from 1, giving $\frac{34}{55}$.

59. There are 15 chocolate bars and 10 children. In how many ways can the chocolate bars be distributed to the children, in each of the following scenarios?

- (a) The chocolate bars are fungible (interchangeable).

We split 15 bars between 10 people, where people can have 0 bars. Using stars and bars, this is $\binom{15+10-1}{15} = \binom{24}{15}$.

- (b) The chocolate bars are fungible, and each child must receive at least one.

Hint: First give each child a chocolate bar, and then decide what to do with the rest.

We imagine lining up 15 chocolate bars. In the 14 spots in between the chocolate bars, we can put a divider. If we place 9 dividers, there are 10 sections and each section has at least one chocolate bar. Furthermore, for all possible ways to sum to 10 through the quantities of chocolate bars for each child, there is a way to place dividers between 15 chocolate bars. The answer is then the number of spots choose the number of dividers, which gives $\binom{14}{9}$.

A way to do it with the hint is to give everyone a chocolate bar. Then we have 5 bars left for 10 children. This is $\binom{5+10-1}{5} = \binom{14}{5} = \binom{14}{9}$.

- (c) The chocolate bars are not fungible (it matters which particular bar goes where).

For each bar, there are 10 choices for who it can go to, so the answer is 10^{15} .

- (d) The chocolate bars are not fungible, and each child must receive at least one.

Hint: The strategy suggested in (b) is does not apply. Instead, consider *randomly* giving the chocolate bars to the children, and apply inclusion-exclusion.

We can count this with the principle of inclusion exclusion. For each S , we count how many ways to assign bars to the children such that the members of S do not receive chocolate.

$$\sum_{S \subset \{1,2,3,4,5,6,7,8,9,10\}} (-1)^{|S|} \cdot (10 - |S|)^{15}$$

$$\sum_{|S|=0}^{10} (-1)^{|S|} \cdot \binom{10}{|S|} \cdot (10 - |S|)^{15}$$

$$\sum_{|S|=0}^{10} (-1)^{|S|} \cdot \binom{10}{10-|S|} \cdot (10-|S|)^{15}$$

$$\sum_{k=0}^{10} (-1)^k \cdot \binom{10}{k} \cdot k^{15}$$

Thus, our answer is $-10+45\cdot 2^{15}-120\cdot 3^{15}+210\cdot 4^{15}-252\cdot 5^{15}+210\cdot 6^{15}-120\cdot 7^{15}+45\cdot 8^{15}-10\cdot 9^{15}+10^{15}$.

60. Given $n \geq 2$ numbers $(a_1, a_2, \dots, a_n)$ with no repetitions, a *bootstrap sample* is a sequence $(x_1, x_2, \dots, x_n)$ formed from the a_j 's by sampling with replacement with equal probabilities. Bootstrap samples arise in a widely used statistical method known as the *bootstrap*. For example, if $n = 2$ and $(a_1, a_2) = (3, 1)$, then the possible bootstrap samples are $(3, 3), (3, 1), (1, 3)$, and $(1, 1)$.

- (a) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$?

For each x_i , there are n possible values, so the answer is n^n .

- (b) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$ if order does not matter (in the sense that it matters only how many times each a_j was chosen, not the order in which they were chosen)?

We can see this is a stars and bars problem, since we have n samples that are distributed over n possible values. The answer is of course $\binom{2n-1}{n}$.

- (c) One random bootstrap sample is chosen (by sampling from $a_1, \dots, a_n$ with replacement, as described above). Show that not all unordered bootstrap samples (in the sense of (b)) are equally likely. Find an unordered bootstrap $\mathbf{b}_1$ that is as likely as possible, and an unordered bootstrap sample $\mathbf{b}_2$ that is as unlikely as possible. Let p_1 be the probability of getting $\mathbf{b}_1$ and p_2 be the probability of getting $\mathbf{b}_2$ (so p_i is the probability of getting the *specific* unordered bootstrap sample $\mathbf{b}_i$). What is $\frac{p_1}{p_2}$? What is the ratio of the probability of getting an unordered bootstrap sample whose probability is p_1 to the probability of getting an unordered sample whose probability is p_2 ?

Since the sample is unordered, rather than representing it as a tuple we can recognize it as $c_1, \dots, c_n$, where c_i is the number of times a_i appears in our sample. The number of ways to draw the bootstrap sample to get c would then be $\binom{n}{c_0} \cdot \binom{n-c_0}{c_1} \cdot \binom{n-c_0-c_1}{c_2} \cdot \dots \cdot \binom{n-c_1-\dots-c_{n-1}}{c_n} = \frac{n!}{c_0! \cdot c_1! \cdot \dots \cdot c_n!}$.

We want to maximize this value, so if we let $c_i = 1$ for all i , then there will be $n!$ ways to draw c , which is the highest possible. This gives a probability of $p_1 = \frac{n!}{n^n}$.

For the smallest probability, we can realize that the solution is to let any one of $c_i = n$ and the rest be 0. This gives only one way to draw c , resulting in a probability of $p_2 = \frac{1}{n^n}$.

Thus, $\frac{p_1}{p_2} = n!$.

We must now find the probability of getting a sample with p_1 , and likewise the probability of getting a sample with p_2 .

For p_1 , it means we need exactly one of each a_i . This means no duplicates, so there are $n!$ ways to sample without dupes, meaning the probability of drawing a sample with probability p_1 is p_1 . This

makes sense, as all samples drawn with probability p_1 are the same sample.

For p_2 , it means we need to draw the same thing each time. We can draw anything for the first value, but subsequently we will need to get exactly the same as the first draw, giving a probability of $\frac{1}{n^{n-1}}$.

So the ratio is $\frac{\frac{n!}{n^{n-1}}}{\frac{1}{n^{n-1}}} = \frac{n!}{n^n} \cdot \frac{n^{n-1}}{1} = (n-1)!$.

61. There are 100 passengers lined up to board an airplane with 100 seats (with each seat assigned to one of the passengers). The first passenger in line decides to sit in a randomly chosen seat (with all seats equally likely). Each subsequent passenger takes their assigned seat if available, and otherwise sits in a random available seat. What is the probability that the last passenger in line gets to sit in their assigned seat? (This is a common interview problem, and a beautiful example of the power of symmetry).

Hint: Call the seat assigned to the j th passenger in line "seat j " (regardless of whether the airline calls it seat 23A or whatever). What are the possibilities for which seats are available to the last passenger in line, and what is the probability of each of those possibilities?

Passengers 2, 3, ..., 99 will ensure that their assigned seat has been sat in. In other words, if it's vacant by their turn, they'll sit in it, and if it's occupied, well, then it was occupied. Thus, if the final passenger's seat is taken, he will sit in passenger 1's seat. It turns out, in any situation where passenger 100 has to sit in passenger 1's seat, there is a corresponding situation of equal probability (all the events are the same, EXCEPT whoever sat in passenger 100's seat chooses to seat in passenger 1's seat) where passenger 100 gets the seat. Thus, the answer is $\frac{1}{2}$.

62. In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $\mathbf{p} = (p_1, p_2, \dots, p_{365})$ be the vector of birthday probabilities, with p_j being the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers).

The k th *elementary symmetric polynomial* in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables. For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$.

Now let $k \geq 2$ be the number of people.

- (a) Find a simple expression for the probability there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.

If p is the probability that all k birthdays are unique, then the answer is $1 - p$.

If $k \leq 365$, then $p = e_k(\mathbf{p}) \cdot k!$. This is because for every set of k birthdays, there are $k!$ ways to assign them to the people.

The answer is $1 - e_k(\mathbf{p}) \cdot k!$

- (b) Explain intuitively why it makes sense that $P(\text{at least one birthday match})$ is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.

I actually struggle to answer this question. It is not that intuitive for me. However, I can see that if not all p_i are equal, then some of them are more likely than others, so the probability of people matching could be higher.

- (c) The famous *arithmetic mean-geometric mean inequality* says that for $x, y \geq 0$,

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x - y)^2 \geq 0$.

Define $\mathbf{r} = (r_1, \dots, r_{365})$ by adding $r_1 = r_2 = \frac{p_1 + p_2}{2}$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$$e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n),$$

show that

$$P(\text{at least one birthday match}|\mathbf{p}) \geq P(\text{at least one birthday match}|\mathbf{r}).$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the "given $\mathbf{r}$ " notation means that the birthday probabilities are given by $\mathbf{r}$. Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

We first show

$$e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n),$$

is true.

For every combination of k from $x_1, \dots, x_n$, they either have both x_1, x_2 , only one, or neither. These sets are disjoint. Thus, we can sum them separately.

The sum of the products which have both x_1, x_2 is $e_{k-2}(x_3, \dots, x_n)$, and we multiply this sum by $x_1 x_2$ to get the sum of all products including $x_1 \cdot x_2$.

To have exactly one of x_1, x_2 means to have $k-1$ of $x_3, \dots, x_n$, so we also add $(x_1 + x_2) e_{k-1}(x_3, \dots, x_n)$.

To have neither of course means to have k of $x_3, \dots, x_n$, giving $e_k(x_3, \dots, x_n)$.

We can indeed see that $e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$.

We return to our formula from part (a) that tells us the probability there is a match, being $1 - e_k(\mathbf{p}) \cdot k!$. We consider using $\mathbf{r}$ instead of $\mathbf{p}$, which would give $1 - e_k(\mathbf{r}) \cdot k!$. However, we know that this is equal to

$$\begin{aligned} e_k(\mathbf{r}) &= e_k\left(\frac{p_1 + p_2}{2}, \frac{p_1 + p_2}{2}, p_3, \dots, p_n\right) \\ &= \frac{(p_1 + p_2)^2}{4} e_{k-2}(p_3, \dots, p_n) + (p_1 + p_2) e_{k-1}(p_3, \dots, p_n) + e_k(p_3, \dots, p_n) \end{aligned}$$

Meanwhile, we can apply the same logic to $e_k(\mathbf{p})$:

$$e_k(\mathbf{p}) = p_1 p_2 e_{k-2}(p_3, \dots, p_n) + (p_1 + p_2) e_{k-1}(x_3, \dots, x_n) + e_k(p_3, \dots, p_n)$$

We can see that $e_k(\mathbf{p}) - e_k(\mathbf{r}) = (p_1 p_2 - \frac{(p_1 + p_2)^2}{4}) e_{k-2}(p_3, \dots, p_n)$.

Since $\frac{p_1 + p_2}{2} \geq \sqrt{p_1 p_2}$ by the arithmetic mean-geometric mean inequality, $(\frac{p_1 + p_2}{2})^2 \geq p_1 p_2$, meaning $e_k(\mathbf{p}) - e_k(\mathbf{r}) \leq 0$ and $e_k(\mathbf{p}) \leq e_k(\mathbf{r})$. Thus, $1 - e_k(\mathbf{p}) \geq 1 - e_k(\mathbf{r})$.

If $(\frac{p_1 + p_2}{2})^2 = p_1 p_2$, then $\frac{p_1^2 + 2p_1 p_2 + p_2^2}{4} - \frac{4p_1 p_2}{4} = 0$, and $(p_1 - p_2)^2 = 0$, implying $p_1 = p_2$. By the contrapositive, if $p_1 \neq p_2$, then $(\frac{p_1 + p_2}{2})^2 \neq p_1 p_2$.

Thus, if $p_1 \neq p_2$, then $p_1 \neq \frac{p_1 + p_2}{2}$ and $p_2 \neq \frac{p_1 + p_2}{2}$, and $(\frac{p_1 + p_2}{2})^2 \neq p_1 p_2$, meaning if $\mathbf{p} \neq \mathbf{r}$ (the inequality here means averaging the two elements changed the vector, which we have already shown means the arithmetic mean-geometric mean inequality is a strict inequality if so).

Thus, we can always average two elements to achieve an answer that's better, unless all the elements are equal, which would ofcourse mean $p_i = \frac{1}{365}$ for all i .